



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION



Major Project Report

THRYLOS INDIA - Admin Management Portal

(TEAM ID : 26E2109)

admin.thrylos.in

Project Category: Industry-Based Project

Submitted in partial fulfilment of the requirement of the degree of

BACHELOR OF TECHNOLOGY (Computer Science and Engineering)

Semester IV — Academic Year 2025–2026

to

K. R. MANGALAM UNIVERSITY, GURUGRAM

Sohna Road, Gurugram, Haryana — 122001

by

S.No	Student Name	Enrollment No.	Programme	Section
1	Utsav Ratan	2401010046	B.Tech CSE Core	B
2	Ishan Jha	2401010022	B.Tech CSE Core	B
3	Rashmi Jha	2401010166	B.Tech CSE Core	B
4	Prem Kumar	2401010243	B.Tech CSE Core	B
5	Vicky Jha	2401010248	B.Tech CSE Core	B
6	Vivek Rai	2401010303	B.Tech CSE Core	B

Under the Supervision of

Internal Mentor: Suman KD Sir

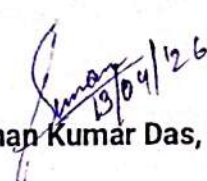
Department of Computer Science and Engineering

School of Engineering and Technology

K. R. Mangalam University, Gurugram — 122001, India

CERTIFICATE

This is to certify that the Project Report entitled, "Thrylos India Admin Management Portal" submitted by "Utsav Ratan (2401010046), Ishan Jha (2401010022), Rashmi Jha (2401010166), Prem Kumar (2401010243), Vicky Jha (2401010248) and Vivek Rai (2401010303)" to K.R. Mangalam University, Gurugram, India, is a record of bonafide project work carried out by them under my supervision and guidance and is worthy of consideration for the partial fulfilment of the degree of Bachelor of Technology in Computer Science and Engineering of the University.


Mr. Suman Kumar Das, Technical Trainer

Date: 12th April 2026

DECLARATION

We hereby declare that this project report titled "Thrylos India Admin Management Portal" is the result of our original work, carried out under the industry mentorship and guidance of Thrylos India, Gurugram. This work has not been submitted elsewhere, in whole or in part, for any other degree, diploma, or academic examination.

The portal described herein is a fully operational, live production system deployed at admin.thrylos.in, serving real users and underpinning the day-to-day internal operations of Thrylos India. All third-party technologies, libraries, frameworks, and APIs used in this project have been duly acknowledged in the Technology Stack section of this report. The information contained herein is accurate, honest, and complete to the best of our collective knowledge.

We understand and acknowledge that any misrepresentation of facts in this declaration or in the project report shall render us liable to academic and disciplinary action as per the regulations of K. R. Mangalam University.

Utsav Ratan (2401010046) B.Tech CSE Core, K. R. Mangalam University Signature: <u>Utsav</u>	Ishan Jha (2401010022) B.Tech CSE Core, K. R. Mangalam University Signature: <u>Ishan</u>
Rashmi Jha (2401010166) B.Tech CSE Core, K. R. Mangalam University Signature: <u>Rashmi</u>	Prem Kumar (2401010243) B.Tech CSE Core, K. R. Mangalam University Signature: <u>Prem</u>
Vicky Jha (2401010248) B.Tech CSE Core, K. R. Mangalam University Signature: <u>Vicky</u>	Vivek Rai (2401010303) B.Tech CSE Core, K. R. Mangalam University Signature: <u>Vivek</u>

Date: April 2026

Place: K. R. Mangalam University, Sohna Road, Gurugram, Haryana — 122001

INDEX

S. No	Content	Page No
1.	Project Completion Certificate	1
2.	Declaration	2
3.	Abstract	3
4.	Chapter 1 — Introduction	4
4.1	Background of the Project	4
4.2	About the Company — Thrylos India	4
4.3	Motivation	5
5.	Chapter 2 — Literature Review & Comparative Analysis	6
5.1	Problem Statement	7
5.2	Gap Analysis	7
6.	Chapter 3 — Objectives	8
7.	Chapter 4 — Tools and Technology Stack	9
7.1	Technology Justification & Alternatives Considered	11
8.	Chapter 5 — Methodology & Development Lifecycle	12
8.1	Overall Architecture	13
8.2	Authentication Architecture & Security Design	14
9.	Chapter 6 — System Modules — Complete Catalogue (36 Modules)	15
9.1	Access Level Definitions	16
9.2	Module-wise Detailed Descriptions	22
10.	Chapter 7 — Implementation & Experimental Setup	23
10.1	Key Technical Highlights	23
10.2	Dashboard Interface & UX Design	24
11.	Chapter 8 — Results and Discussion	25
11.1	Student Team & Individual Contributions	26
12.	Chapter 9 — Conclusion & Future Work	27
13.	Chapter 10 — Challenges Faced & Solutions Implemented	29
14.	Learning Outcomes & Skill Acquisition	30
15.	Annexure I — Plagiarism Declaration Certificate	31
16.	Reference	32

ABSTRACT

The Thrylos India Admin Management Portal, accessible at admin.thrylos.in, is a fully deployed, production-grade, multi-module enterprise operations platform built entirely by a team of six B.Tech Computer Science and Engineering students from K. R. Mangalam University, Gurugram. Developed over five months from January to April 2026, the portal now serves live users and underpins every critical aspect of Thrylos India's internal operations — spanning human resources management, employee records, esports tournament administration, financial analytics, and real-time team communication.

Built on a modern, industry-standard technology stack comprising React 18, TypeScript, Supabase, Tailwind CSS, Framer Motion, and Recharts, the system demonstrates genuine mastery of full-cycle web application development across 36 fully operational modules organized under four distinct role-based access levels. What sets this project apart from typical academic submissions is its nature: it is not a prototype or a simulation — it is a live, working system currently processing real business operations, real user accounts, and real-time team coordination at Thrylos India every day.

The authentication architecture employs OTP-based two-factor verification with geolocation session logging through the Brevo transactional email API and Supabase Edge Functions running on the Deno runtime, meeting enterprise security requirements. The database layer is protected by PostgreSQL Row-Level Security policies enforced at the engine level, ensuring tamper-proof access control independent of client-side state. All 36 modules are independently deployable, role-aware, and built to production-quality engineering standards with full TypeScript strict-mode compliance.

This report documents the complete project lifecycle from initial requirements gathering through architecture design, phased development, integration, testing, production deployment, and post-deployment operations. It covers the technology decisions and their rationale, the complete module catalogue with detailed functional descriptions, the team's individual contributions, learning outcomes, challenges encountered and resolved, and a roadmap of planned enhancements.

Chapter 1

Introduction

1.1 Background of the Project

Enterprises and technology companies operating in fast-paced, distributed environments increasingly depend on consolidated digital operations platforms to coordinate their teams, manage their human resources, track financial flows, and maintain quality and accountability. The absence of such a unified platform forces organizations to rely on a fragmented combination of tools — spreadsheets for data, messaging apps for communication, email chains for approvals, and disconnected dashboards for reporting — each of which introduces inconsistency, delays, and a lack of cross-functional transparency.

Thrylos India, a registered digital technology company based in Gurugram, Haryana, faced precisely this challenge. As the company expanded its operations across four principal business verticals — esports infrastructure and tournament management, custom web application development, AI-powered tool integration, and enterprise technology services — its internal coordination mechanisms struggled to keep pace with growth. Data was inconsistently maintained across tools, cross-team visibility was poor, and there was no authoritative single source of truth for operational metrics, HR records, or financial performance.

The student project team was engaged by Thrylos India to address this operational challenge comprehensively by designing, building, and deploying a unified, purpose-built internal operations platform that would consolidate every function under one authenticated, role-aware dashboard. This engagement gave the student team the rare opportunity to take complete ownership of a software product from the first line of architecture to a live, production-ready deployment serving real users.

1.2 About the Company — Thrylos India

Thrylos India is a registered digital technology company operating primarily in the Indian market. The company's work sits at the intersection of gaming culture and enterprise technology, providing tournament hosting infrastructure, team management systems, and digital services to esports organizations, educational institutions, and corporate clients across India.

The company maintains a live consumer-facing platform at www.thrylos.in, which handles tournament registrations, QR-based UPI payment integration, and user-facing content. The

admin portal at admin.thrylos.in, built during this project, serves as the company's internal operations backbone, giving its team the tools to manage everything from employee records and leave approvals to financial analytics and tournament player registries from a single, secure interface.

Company Name	Thrylos India
Type of Company	Registered Digital Technology Company
Business Verticals	Esports Infrastructure, Web Development, AI Integration, Tech Services
Public Platform	www.thrylos.in
Admin Portal	admin.thrylos.in
Tech Stack (Public)	React, Node.js, Supabase, QR-based UPI Payment Integration
Headquarters	Gurugram, Haryana, India
Target Market	Esports Organizations, Educational Institutions, Corporate Clients across India

1.3 Motivation

The motivation for this project emerged directly from the operational realities of running a growing technology company with a distributed team. Before the portal was built, Thrylos India managed attendance via manual records, tracked employee information in spreadsheets, communicated internally through third-party messaging applications that offered no custom role-based visibility, handled leave approvals through email chains, and had no centralized mechanism for monitoring team performance or financial data in real time.

As the team grew and operations diversified across esports, development, content creation, and client services, this patchwork of tools became increasingly untenable. Onboarding new team members took longer because there was no central directory or documented operational structure. Financial oversight was difficult because data lived across separate tools. Accountability was hard to enforce because there was no immutable audit trail of who did what and when.

Beyond the operational need, the project offered an extraordinary learning motivation for the student team: the opportunity to build not an academic exercise, but a live system with real users, real consequences, and real feedback. The requirement to deliver a platform that would actually be used by Thrylos India's team every day created a quality standard fundamentally different from a typical laboratory project, pushing every team member to think and work like a professional software engineer.

Chapter 2

Literature Review and Comparative Analysis

2.1 Review of Existing Systems and Related Work

Enterprise operations platforms and internal admin portals represent a well-established category of software, with commercially available solutions including SAP SuccessFactors, Microsoft SharePoint, Zoho People, and numerous HRMS (Human Resource Management System) platforms. Academic literature on enterprise web application development has extensively documented the evolution from monolithic server-rendered applications to component-based single-page applications (SPAs) with real-time backend services.

Work by Nielsen Norman Group and others on enterprise UX has highlighted that the primary usability failure of internal tools is cognitive overload — exposing more information than users need to act, without progressive disclosure or role-based filtering. This directly informed the portal's module card system, which surfaces only the key operational metrics for each module on the dashboard rather than opening modules by default.

Research into real-time web applications using WebSocket infrastructure, particularly the seminal work around Pusher, Firebase Realtime Database, and later Supabase Realtime, established that pub-sub WebSocket architectures are the appropriate solution for multi-user collaborative tools requiring instant data synchronization — as opposed to polling approaches which introduce latency and server load proportional to user count. This literature directly justified the choice of Supabase Realtime for the Team Chat, Notification Center, and Task Board modules.

Security literature around two-factor authentication in enterprise applications — particularly NIST SP 800-63B guidelines for digital identity — informed the decision to implement OTP-based second-factor authentication delivered over verified transactional email rather than SMS, which is vulnerable to SIM-swapping attacks. The implementation of geolocation session logging goes beyond minimum NIST recommendations and reflects emerging best practices from zero-trust security architectures.

2.2 Gap Analysis

Commercial off-the-shelf platforms were evaluated during the discovery phase but found unsuitable for several key reasons. Generic HRMS platforms do not natively accommodate the esports-specific operational needs of Thrylos India — there is no standard HRMS module for

player registries, tournament revenue tracking, or gaming-handle management. Extending these platforms to cover such domains would require expensive customization work exceeding the complexity of building a purpose-built solution.

Commercially available chat and collaboration tools (Slack, Microsoft Teams) provide communication features but do not integrate with the company's HR data, performance scoring engine, or financial analytics — meaning the team would still require separate tools for these functions. Integration costs and the complexity of maintaining authenticated sessions across multiple platforms introduced more overhead than a unified solution.

The gap identified was the absence of a single, integrated, role-aware operations platform that would unify HR management, real-time communication, performance tracking, financial analytics, esports operations, and enterprise security in one authenticated dashboard, purpose-built for the specific operational context of Thrylos India. This portal fills that gap.

2.3 Problem Statement

The closed absence of a unified internal operations platform at Thrylos India created the following measurable operational problems: employee attendance was tracked inconsistently across different tools with no exportable audit trail; leave approvals required multiple email exchanges and had no structured workflow or leave balance tracking; there was no single source of truth for employee records including KYC data, salary structures, and department assignments; team communication was fragmented across external tools with no role-based visibility control; financial performance and esports revenue data were not consolidated and not accessible in real time; and there was no mechanism to log, audit, or review admin actions across the organization for security and accountability purposes.

The problem to be solved was therefore: design and deploy a production-grade, unified internal operations platform that consolidates all of the above functions under one authenticated, role-aware interface, delivered as a live system serving real users rather than as a prototype or proof-of-concept.

Chapter 3

Objectives

The following objectives were defined at the project outset in collaboration with Thrylos India's management team and used to measure the success of the engagement throughout the development lifecycle:

1. Design and deploy a secure, role-based web application as the company's single internal operations platform, accessible from any browser without additional software installation.
2. Implement enterprise-grade authentication using OTP-based two-factor verification delivered over verified transactional email, with comprehensive geolocation session logging capturing GPS coordinates, IP-layer location, ISP, and session timestamps.
3. Develop a minimum of 30 production-ready modules covering all operational domains including HR, finance, analytics, communication, data management, and esports operations, with each module serving real functional use cases.
4. Integrate real-time capabilities for team chat, live notifications, attendance feeds, task management, and performance score updates using WebSocket-based pub-sub architecture.
5. Apply industry-standard development practices throughout: TypeScript strict mode for the entire codebase, component architecture with reusable design patterns, database Row-Level Security policies at the PostgreSQL engine layer, and CI/CD deployment pipelines enabling zero-downtime releases.
6. Deliver a live, production-hosted system serving actual Thrylos India users — not a prototype, demonstration, or simulation — with the domain `admin.thrylos.in` configured and operational.
7. Implement role-based access control with at least three distinct role levels, with access boundaries enforced both at the React application layer (route-level protection) and at the Supabase database layer (RLS policies), ensuring defence in depth.
8. Document the system comprehensively, including a GitHub repository with full module catalogue, authentication flow diagrams, API documentation, and this formal academic project report.

All eight objectives were met upon delivery. The final system contains 36 production-ready modules — six more than the minimum specified — with four distinct role levels rather than three, a fully live production deployment at `admin.thrylos.in`, and documentation covering every aspect of the system from architecture to individual module features.

Chapter 4

Tools, Platform, and Technology Stack

Every technology in the stack was selected according to three criteria applied consistently: industry adoption (the technology is actively used in production by leading technology companies, not just academically studied), developer productivity (excellent documentation, strong tooling ecosystem, and availability of community-supported component libraries), and long-term scalability (the ability to handle growing data volumes, expanding user bases, and increasing module counts without requiring architectural rewrites).

4.1 Frontend Framework — React 18 with TypeScript

React 18 was selected as the foundation of the frontend application for its component-based architecture, which is the ideal paradigm for a large, multi-module SPA requiring 36 independently navigable screens. The component model enables strict encapsulation of module-specific logic, making each module independently testable and deployable without affecting others.

TypeScript was applied in strict mode across the entire codebase — a deliberate decision to enforce compile-time type safety throughout a project that would eventually comprise over 50,000 lines of code written by six different developers. TypeScript's discriminated union types, generics, and strict null-check mode caught hundreds of potential runtime errors during development, particularly at integration boundaries between modules that shared data types such as `User`, `Admin`, `Task`, `Leave`, and `AttendanceRecord`.

4.2 Styling System — Tailwind CSS with shadcn/ui

Tailwind CSS was chosen over alternatives like Bootstrap and Material UI for its utility-first approach, which allows pixel-precise custom designs directly in JSX without fighting against opinionated pre-styled component defaults. For a portal with a distinctive visual identity — a dark header, branded colour scheme, animated card grids, and a consistent information density across all modules — Tailwind's flexibility was essential.

shadcn/ui was integrated as the accessible component primitive layer, providing ARIA-compliant dialogs, dropdowns, tables, popovers, and form controls built on Radix UI. These components handle keyboard navigation, focus trapping, and screen reader compatibility out of the box, ensuring the portal meets basic accessibility standards without requiring manual ARIA implementation for every interactive element.

4.3 Animations — Framer Motion

Framer Motion was selected as the animation library for its production-quality React-first API that goes significantly beyond CSS transitions. The portal uses Framer Motion for page transition animations (fade + slide between modules), staggered card entrance sequences on the dashboard (each of the 36 module cards animates in with a cascade delay, creating the perception of fast loading even when data is being fetched), layout animations on dynamic lists, and gesture-driven interactions. The animation system was central to achieving the high-quality, professional feel of the interface that distinguishes it from a typical academic project.

4.4 Backend and Database — Supabase (PostgreSQL)

Supabase was selected as the backend platform because it provides a fully managed PostgreSQL database with the critical feature of Row-Level Security — access control enforced at the database engine level, not the application level. This architectural property is what enables the portal's defence-in-depth security model: even if an attacker bypassed the React frontend entirely and queried the Supabase REST API directly with a stolen JWT token, they would only receive data their authenticated role permits.

Supabase also provides auto-generated REST and real-time APIs, serverless Edge Functions for custom server-side logic, file storage with access-controlled bucket policies, and a generous free tier suitable for the project's scale. This combination replaced what would otherwise require a separate Node.js API server, a separate database server, and a separate file storage service — significantly simplifying the deployment architecture and reducing operational overhead.

4.5 Authentication — OTP via Brevo API and Supabase Edge Functions

The authentication system was built on Supabase Edge Functions running on the Deno TypeScript runtime rather than within the React client. This is a fundamental security architecture decision: OTP generation, email dispatch, and quota enforcement logic must run server-side. If any part of the OTP mechanism ran in the browser, a technically sophisticated attacker could extract the OTP generation algorithm, bypass the email delivery requirement, and authenticate without controlling the registered email address.

Brevo (formerly Sendinblue) was selected as the transactional email provider for its reliable free tier, straightforward API, and easy DKIM and SPF domain verification workflow. Emails are sent from the @admin.thrylos.in domain, which is verified in Brevo with proper DNS records, ensuring OTP emails consistently reach inboxes rather than spam folders.

4.6 Real-Time Engine — Supabase Realtime

Supabase Realtime provides a WebSocket-based publish-subscribe infrastructure that enables live data synchronization across all connected clients without polling. This powers five modules that require instant multi-user updates: the Team Chat module (where messages must appear instantly for all participants), the Notification Center (where system events must push to all eligible recipients in real time), the Task Board (where card movements and status changes must be visible to all assignees), the Daily Standups feed (where new entries must appear for managers as they are submitted), and the Performance Scores leaderboard (which recalculates in real time as new logs are submitted).

4.7 Data Visualization — Recharts

Recharts was chosen for its React-first API built on D3.js, providing composable chart primitives that integrate naturally with React's component and state model. It was used across the Analytics Dashboard, Social Media Analytics, Performance Scores, and HR Dashboard for bar charts, line charts, area charts, pie charts, and composite multi-axis graphs with dual Y-axes. All charts use responsive containers that reflow to fit any screen width, hover tooltips showing exact data point values, and accessible ARIA labels.

4.8 Deployment — Vercel with CI/CD

Vercel was selected for frontend deployment because of its zero-configuration React deployment workflow, automatic preview deployments for every pull request (enabling the team to review features before merging), global CDN edge network for fast load times across India, and seamless custom domain configuration. The CI/CD pipeline is configured so that every push to the main branch on GitHub triggers an automatic production build and deployment. Combined with Supabase Cloud hosting the database, Edge Functions, storage, and real-time engine with a 99.9% uptime SLA, this creates a fully managed, highly available production infrastructure requiring no manual server administration.

4.9 Technology Choice Justification

React with TypeScript was preferred over Vue.js and Angular because React's ecosystem maturity, TypeScript's compile-time safety across a large multi-developer codebase, and the team's existing foundational knowledge made it the most productive choice. Supabase was preferred over Firebase because Supabase's PostgreSQL foundation provides relational data integrity critical for HR, finance, and audit data — the kind of structured, relationally linked data that does not fit naturally in Firebase's document-based NoSQL model, and where Row-Level Security policies provide access control that Firebase cannot match. Tailwind CSS was preferred over Bootstrap and MUI because of its utility-first flexibility for custom designs. Vercel was preferred over AWS and Railway for its zero-config React deployment and automatic preview environments.

Chapter 5

Methodology and Development Lifecycle

5.1 Overall Architecture

The portal follows a modern single-page application architecture with a clear separation between the frontend presentation layer and the backend data/services layer. The React 18 TypeScript application runs entirely in the browser and communicates with Supabase exclusively over two channels: the Supabase JavaScript client library (which provides authenticated REST API access to PostgreSQL tables respecting RLS policies) and Supabase Realtime WebSocket subscriptions (which deliver live data changes to subscribed clients).

All sensitive operations — OTP generation, email dispatch, and quota enforcement — run in Supabase Edge Functions on the Deno runtime, isolated from the browser entirely. File assets are stored in Supabase Storage buckets with access policies tied to RLS roles. The frontend is deployed on Vercel's global CDN edge network, with all environment variables (Supabase URL, anon key) managed as Vercel project secrets rather than in source code. There is no traditional Node.js or Express.js API server in the stack — Supabase's auto-generated REST API, augmented by Edge Functions for custom logic, replaces it entirely.

5.2 Authentication Architecture and Security Design

The authentication system was designed as a multi-layer security mechanism that goes significantly beyond standard username-and-password login. The design goal was to ensure that every login to the admin portal is verified through a second factor, physically location-stamped, and permanently logged, creating a complete chain of custody for every access event.

The authentication flow proceeds through nine distinct steps. First, the admin submits their registered email address through a React form with client-side format validation; unregistered emails are rejected immediately without any server call, preventing unnecessary Edge Function invocations. Second, a Supabase Edge Function (Deno runtime) generates a cryptographically random 6-digit OTP and stores it alongside an expiry timestamp in a secure Supabase table. Third, the same Edge Function calls the Brevo transactional email API to deliver the OTP to the admin's registered email address from the @admin.thrylos.in verified sender domain. Fourth, the system checks the admin's daily OTP request count against a configurable limit stored in Supabase, rejecting requests exceeding the quota to prevent brute-force fishing and control API costs.

Fifth, the admin submits the 6-digit OTP; the Edge Function validates it against the stored hash, checks the expiry window (typically 10 minutes), and marks the OTP as consumed to prevent replay attacks. Sixth, on successful OTP verification, the browser's Geolocation API requests GPS coordinates from the user, which are required for portal access. Seventh, the client sends its public IP address to a

geolocation microservice that returns city, region, country, ISP, and network type. Eighth, a complete session record including email, timestamp, GPS coordinates, IP address, city, ISP, and country is written to the Audit Logs table as an immutable record. Ninth, the authentication context loads the admin's role from Supabase and the React router renders only the modules authorized for that role.

Beyond the application-layer role checks, the database is protected by Supabase Row-Level Security policies applied directly at the PostgreSQL table level. A Regular Admin can only read their own attendance and work log records. An HR Admin can read all employee records but cannot access financial audit logs. Only Super Admins have unrestricted read and write access across all tables. These policies are enforced at the database engine level, making them tamper-proof regardless of client-side state or JWT manipulation.

5.3 Project Timeline and Milestones

The project was executed over five months, from January to April 2026, in eight structured phases with clear deliverables per milestone:

January 2026 — Kickoff Phase: Team formation, requirements gathering through stakeholder interviews with Thylos India management, Supabase project setup, base GitHub repository creation with folder structure and branch protection policies, and authentication flow design. The module requirement list was finalized at 36 modules during this phase.

January–February 2026 — Architecture Phase: PostgreSQL database schema design with all table relationships normalized to 3NF, RLS policy authoring for all tables, component library configuration with shadcn/ui and Tailwind theme tokens, React Router v6 routing architecture with protected route wrappers, and role matrix definition mapping each of the 36 modules to its access level.

January 2026 — Phase 1 Development: Core foundation modules built and integrated: Dashboard (Overview Stats), Attendance Tracker, Admin Management, Audit Logs, Certificate Manager, Employee Management, and Leave Management. These seven modules formed the operational backbone upon which all subsequent modules depended.

January–February 2026 — Phase 2 Development: Communication and collaboration modules completed: Real-Time Team Chat with Supabase Realtime WebSocket infrastructure, Notification Center, Announcements board, Feedback channel, Polls and Surveys, Task Board (Kanban), Daily Standups, and Team Events calendar.

February 2026 — Phase 3 Development: Analytics and reporting modules completed: Analytics Dashboard with multi-axis Recharts visualizations, Social Media Analytics, Performance Scores engine with configurable weighted metrics, Admin Reports, and Overview Stats enhancements.

February–March 2026 — Phase 4 Development: Operations and productivity modules completed: Esports Players registry, HR Dashboard, Internship Tracker, Expense Claims, Bulk Upload and Import with CSV validation, File and Media Manager, Career Applications, Holiday Calendar, Birthday Reminders, Email Management, Quick Notes, Company Directory, Meeting Notes, and Company Links.

March 2026 — Integration Phase: Full system integration connecting all 36 modules to their respective Supabase data pipelines, Realtime subscription wiring for modules requiring live updates, cross-module data flows (e.g., Performance Scores pulling from Attendance, Work Logs, Task Board, and Standups simultaneously), and performance optimization through query caching and component memoization.

March–April 2026 — Testing and Deployment: End-to-end functional testing of all 36 modules, cross-browser compatibility testing on Chrome, Firefox, Edge, and Safari, mobile responsiveness validation on multiple device form factors, security review of RLS policies and Edge Function logic, production deployment to admin.thrylos.in via Vercel with CI/CD pipeline configuration, custom domain DNS setup, and environment variable management for production secrets. Documentation was finalized during this phase.

5.4 Scope of Work

The project scope encompassed the complete software development lifecycle from initial requirements gathering through post-deployment operations and documentation. Discovery included stakeholder interviews, existing workflow analysis, and a formal module specification document. Architecture covered system design, database schema, RLS policy planning, component hierarchy mapping, and the role matrix definition. UI/UX Design produced wireframes, the component library setup, design token definitions, and animation specifications. Frontend Development produced all 36 React TypeScript module implementations plus global routing, authentication context, role-aware rendering, and the responsive layout system. Backend Development covered the PostgreSQL schema, all RLS policies, Supabase Edge Functions for OTP and email, the Brevo API integration, and storage bucket configuration. Integration connected all modules through Supabase Realtime subscriptions, API hooks, state management, and error handling. Testing covered manual functional testing of all modules, cross-browser and mobile validation, and a security review. Deployment configured the Vercel production pipeline with the custom admin.thrylos.in domain. Documentation produced the GitHub README with module catalogue, auth flow diagrams, API documentation, and this project report.

Chapter 6

System Modules — Complete Catalogue

6.1 Access Level Definitions

The portal implements four access levels that control which modules are visible and which operations are permitted for each authenticated admin. These access boundaries are enforced at two independent layers: the React application layer (routes are protected and module cards are conditionally rendered based on the authenticated role from context) and the Supabase database layer (Row-Level Security policies restrict data access at the PostgreSQL engine level independent of application logic).

Super Admin is assigned to founders, directors, and senior management and provides full unrestricted access to all 36 modules including sensitive financial, HR, and security modules. HR Admin is assigned to human resources personnel and provides access to all HR-related modules: Employee Management, Leave Management, HR Dashboard, Internship Tracker, and Certificate Manager. All Users applies to any authenticated admin regardless of role level and covers approximately 22 of the 36 modules including all operational and communication modules that do not require elevated privilege. Modules carrying a Live badge are modules that display real-time data feeds; this badge is informational and does not constitute an additional access requirement.

6.2 Module Directory — All 36 Modules with Detailed Descriptions

Module 1: Overview Stats (All Users — Dashboard & Analytics)

The Overview Stats module is the central dashboard entry point of the entire portal and the first screen any authenticated admin encounters after login. It provides a real-time bird's-eye view of Thrylos India's complete operations, with role-specific content rendering so that each admin sees information appropriate to their responsibilities. The dashboard is divided into five horizontal zones: a personalized welcome banner with the admin's display name and live activity status, a Today's Tasks widget with an inline task creator and mood selector, a Super Admin accordion panel with live attendance statistics and recent work logs (visible only to Super Admins), a statistics row showing four live-fetched cards (Total Users, Esports Users, Average Followers, Employees), and the full 36-module card grid with staggered Framer Motion entrance animations. Every metric card on the dashboard is connected to a Supabase Realtime subscription and updates automatically as underlying data changes without requiring a page refresh.

Module 2: Attendance Tracker (All Users — HR & Operations)

The Attendance Tracker is a comprehensive daily attendance management system covering all admins and employees. It allows marking attendance with a single click, recording absences with categorized reasons including sick leave, work from home, field visit, and personal leave, viewing weekly and monthly attendance summaries per individual, and exporting attendance reports in CSV format for payroll processing or HR review. The module integrates directly with the Performance Scores engine, which reads attendance percentage as one of its weighted scoring inputs, and with the Holiday Calendar, which automatically excludes designated public

holidays from working-day calculations. Super Admins can view the full attendance record for all team members; regular admins can only view and update their own attendance in compliance with RLS policies.

Module 3: Real-Time Team Chat (All Users – Communication)

The Real-Time Team Chat module is the most technically complex component in the portal and the one that most directly replaces an external dependency. Built entirely on Supabase Realtime's WebSocket pub-sub infrastructure, it maintains persistent two-way connections between all connected clients so that messages appear instantly without any polling mechanism. The module supports multiple concurrent conversation threads, file and image attachment uploads stored in Supabase Storage with access-controlled signed URLs, full message reply threading with visual parent-message linking, keyword-based message search with highlighted results, per-conversation unread message counters, and real-time online presence indicators showing which team members are currently active. The layout is fully mobile-responsive with a collapsible conversation list sidebar. Building this module from scratch rather than integrating a third-party SDK was a deliberate choice to keep all message data within the company's own Supabase database, ensuring full control over data, security, and compliance.

Module 4: Analytics Dashboard (Super Admin – Dashboard & Analytics)

The Analytics Dashboard is an advanced business intelligence module providing deep financial and operational insights through interactive Recharts-powered visualizations. It features a composite multi-axis chart showing revenue trends as a bar chart overlaid with monthly growth rate as a line chart on a secondary Y-axis, domain-wise revenue breakdown comparisons, total revenue summaries, and period-over-period growth rate indicators. All charts support hover tooltips with exact data values, responsive containers that reflow to any screen width, and live updates through Supabase Realtime subscriptions. This module is restricted to Super Admins because it exposes company-level financial data that should not be visible to operational staff.

Module 5: Payment Verification (All Users – Finance)

The Payment Verification module handles the verification and management of all user payment transactions received through the Thrylos India platform. Admins can view pending transactions with full metadata, verify transaction IDs against payment gateway records, mark transactions as verified or disputed with reason logging, export payment records in CSV or PDF formats, and track daily verification counts with a running verified-today counter. The module is essential for Thrylos India's esports tournament registration business, where hundreds of payment transactions may need verification in a single day.

Module 6: Certificate Manager (Super Admin – HR & Operations)

The Certificate Manager provides an end-to-end digital certificate issuance and verification platform. Certificates can be issued for interns, event participants, course completions, tournament finalists, and any other milestone the company recognizes. Each certificate receives a cryptographically unique ID that can be verified by third parties through a public-facing verification URL, providing tamper-proof proof of issuance that recipients can share with universities or employers. The module tracks total certificates issued and this-month counts with full issuance history including the issuing admin, recipient name, certificate type, and timestamp. Integration with the Internship Tracker enables automatic certificate issuance on internship completion.

Module 7: Internship Tracker (Super Admin – HR & Operations)

The Internship Tracker covers the complete intern lifecycle from initial onboarding through active management to final exit and certificate issuance. It tracks each active intern's assigned tasks with deadlines, daily attendance (separate from the main Attendance Tracker to allow interns who are not full admin accounts to be tracked), performance ratings updated by their assigned mentors, and certificate eligibility based on configurable completion criteria. Managers can monitor the real-time count of active interns, total completed internships, and individual progress dashboards. The module integrates with the Certificate Manager for seamless certificate issuance on verified internship completion.

Module 8: Admin Management (Super Admin – System Administration)

Admin Management is the core control panel for managing all admin accounts within the system. Super Admins can create new admin accounts with assigned role levels, view complete activity logs per admin including login timestamps and modules accessed, manage per-admin OTP request quotas to control Brevo API usage, reset admin credentials through the Edge Function authentication flow, and deactivate accounts for team members who leave the organization. The module displays total admin count and currently active admin count. All changes made through this module are logged in the Audit Logs module for compliance and accountability.

Module 9: Bulk Upload and Import (Super Admin – Data Management)

The Bulk Upload and Import module addresses the operational need to load large datasets without direct database access. It accepts CSV and Excel files for tournament player registrations, employee onboarding batches, inventory updates, and other structured data imports. The module parses files client-side using the Papa Parse library and validates every row against predefined schema rules covering required fields, data types, and format constraints before any data reaches Supabase. The validation report highlights failing rows with specific field-level error messages, allows the user to download a corrected template, and shows an overall success rate percentage. Only validated rows are submitted on final confirmation. For large files, the module uses Supabase's bulk upsert API, which was essential for handling 500+ row imports within Supabase's connection pool constraints.

Module 10: Notification Center (All Users – Communication)

The Notification Center is a centralized, real-time alert hub aggregating notifications from across the entire platform. It delivers instant alerts for new payment orders requiring verification, match results from esports tournaments, stock changes, task assignments and updates, leave approval decisions, system events like new admin account creation, and any other cross-module events configured to trigger notifications. Role-based notification filtering ensures each admin only sees alerts relevant to their responsibilities. The module tracks unread notification count (displayed as a badge on the navigation icon) and today's total notification count.

Module 11: File and Media Manager (All Users – Data Management)

The File and Media Manager provides cloud storage management for all binary assets used across the organization. Admins can upload, organize into folders, preview, and generate shareable download links for tournament posters, event certificates, financial receipts, employee KYC documents, and general company files. All files are stored in Supabase Storage buckets with access policies tied to RLS roles, ensuring that sensitive HR documents are only downloadable by HR Admin and Super Admin roles. The module tracks total file count and current storage usage and supports folder organization for logical asset grouping.

Module 12: Audit Logs (Super Admin – Security & Compliance)

The Audit Logs module is the foundational security and compliance infrastructure of the entire portal. It permanently records every significant admin action across the system: login events with full geolocation data, module access timestamps, data record creation, modification and deletion events, certificate issuances, report exports, configuration changes, and failed authentication attempts. Each record captures the admin's identity and role, action type, affected module and record ID, timestamp with millisecond precision, public IP address, geolocation (city, region, country, ISP), and GPS coordinates. Records are insert-only at the RLS level — no admin, including Super Admins, can delete or modify audit entries, making the log forensically reliable. The module provides a filterable, searchable interface with filters for admin, date range, action type, and module, plus CSV export.

Module 13: Employee Management (Super Admin – HR & Operations)

Employee Management is the comprehensive HR records system for all company employees. It stores and manages complete employee profiles including personal details, Aadhaar card numbers, PAN card numbers, bank account information for payroll, designation, department, reporting manager, joining date, salary structure, and employment status (active, on leave, resigned, terminated). Document uploads per employee support KYC document storage with Supabase Storage integration. The module supports search by name, department filter, and designation filter, enabling HR administrators to quickly locate specific employee records in a growing organization.

Module 14: Career Applications (Super Admin – HR & Operations)

The Career Applications module tracks all job applications received through the Thyrylos India careers portal. Reviewers can access complete applicant profiles including submitted CV/portfolios, update application status through a configurable hiring pipeline (Applied → Screening → Interview Scheduled → Interview Completed → Offered → Hired or Rejected), add internal notes visible only to HR and Super Admin, and track total application count and pending review backlog. The module provides a structured workflow replacing email-based recruitment tracking.

Module 15: Leave Management (All Users – HR & Operations)

The Leave Management module provides a structured leave request and approval workflow for all admins and employees. Users submit leave applications specifying leave type (casual, sick, earned, compensatory, or unpaid), the date range, and the reason. Requests flow through a configurable approval chain ending with the authorized approver (typically an HR Admin or Super Admin). The module tracks total leave requests submitted, pending approvals, leave balance per employee with automatic deduction on approval, and full leave history per employee. Integration with the Holiday Calendar ensures that public holidays within a requested leave date range are automatically excluded from the leave balance deduction.

Module 16: HR Dashboard (HR Admin – HR & Operations)

The HR Dashboard is a dedicated, high-level operations hub designed exclusively for the HR Admin role, consolidating all human resource management functions into a single interface. It provides summary views of total employee headcount, active intern count, leave requests pending approval, certificates issued this month, upcoming birthdays within the next 14 days, and open job applications in the hiring pipeline. Quick-action

shortcuts directly into Employee Management, Leave Management, Internship Tracker, and Certificate Manager, reduce navigation overhead for HR personnel who need to move between these modules many times daily.

Module 17: Holiday Calendar (Super Admin – HR & Operations)

The Holiday Calendar provides company-wide holiday schedule management. Super Admins can add, edit, or remove company-designated holidays with name, date, and optional description. The holiday list is integrated with the Attendance Tracker (holidays are automatically marked as non-working days), the Leave Management module (holiday days within approved leave are not deducted from leave balance), and the Performance Scores engine (performance metrics are calculated against working days only, not calendar days). The module displays total holidays for the current year and provides a visual calendar view.

Module 18: Announcements (All Users – Communication)

The Announcements module provides a structured internal communications board for company-wide and role-targeted official announcements. Authorized admins can post announcements with a title, body text, category (operational, policy, celebration, alert, or general), and target audience (all roles, Super Admin only, or HR Admin and above). Posts are displayed in reverse-chronological order with read and unread indicators per team member. The module replaces email group announcements that previously had no read-tracking, no role targeting, and no categorization.

Module 19: Polls and Surveys (All Users – Communication)

The Polls and Surveys module enables structured team feedback collection and data-driven decision making. Admins can create single-choice or multiple-choice polls with configurable option sets, set a poll duration after which voting closes, and choose between anonymous mode (votes are recorded without attribution) and named mode (visible to Super Admins for accountability). Results display as live percentage bars updated in real time as votes are cast. The module tracks total votes received and participation rate per poll, with full poll history for longitudinal reference.

Module 20: Task Board (All Users – Project Management)

The Task Board is a Kanban-style project and task management system providing full visibility into work in progress across the entire team. Tasks are created with title, description, assignee, priority level (low, medium, high, or critical), due date, and category. Cards move across four configurable columns representing workflow stages: To Do, In Progress, Under Review, and Done. All card movements and status changes are pushed via Supabase Realtime to all team members simultaneously, ensuring everyone sees the current state of the team's work without refreshing. Overdue tasks are automatically flagged with a visual indicator.

Module 21: Daily Standups (All Users – Project Management)

The Daily Standups module provides a structured daily status update mechanism inspired by agile standup meetings. Each admin submits a standup covering three standard fields: what they completed yesterday, what they plan to accomplish today, and any blockers impeding their progress. An optional mood indicator allows team members to signal their current wellbeing. Standups are aggregated in a team feed visible to managers, who can quickly scan the feed to spot blockers requiring intervention and track day-to-day team momentum. Standup consistency data feeds into the Performance Scores engine as one of its weighted scoring metrics.

Module 22: Feedback (All Users – Communication)

The Feedback module provides an open, bidirectional channel for all team members to share suggestions, raise concerns, report bugs in the portal or other company systems, and provide general operational feedback. Both anonymous and named submission modes are available, allowing team members to raise sensitive concerns without attribution if needed. Feedback items are categorized (bug report, feature request, process improvement, recognition, or general) and managed through a resolution workflow: each item can be marked as Under Review, In Progress, Resolved, or Closed with an optional resolution note. This provides accountability for feedback without letting it disappear unacknowledged.

Module 23: Team Events (All Users – Project Management)

The Team Events module provides a shared team calendar for scheduling and tracking all company events including meetings, training sessions, client calls, celebrations, workshops, and project milestones. Events are assigned a type, tagged with relevant attendees from the Company Directory, and configured with optional reminders. The module provides both a calendar grid view for month-level event visibility and a chronological list view of upcoming events sorted by date. Integration with the Notification Center pushes event reminders to relevant attendees automatically.

Module 24: Performance Scores (All Users – Dashboard & Analytics)

The Performance Scores module is an automated, objective performance ranking and scoring engine that calculates individual performance scores for all admins and employees based on configurable weighted metrics. The scoring formula aggregates inputs from six data sources: attendance percentage from the Attendance Tracker, daily work log hours from the Tech Work Dashboard, bug and feature counts from structured work logs, content creation counts from the Content Work Dashboard, task completion rates from the Task Board, and standup submission consistency from Daily Standups. Super Admins can adjust the weight assigned to each metric to tune the model for different team roles. Scores recalculate in real time whenever new data is submitted anywhere in the system. The module displays a comparative leaderboard, individual score breakdowns by metric, and historical trend charts.

Module 25: Admin Reports (Super Admin – Dashboard & Analytics)

Admin Reports enables Super Admins to generate comprehensive per-admin performance and activity reports aggregating data from across the portal over any selectable date range. Reports compile attendance records, daily work logs, leave history, task completion rates, standup submission counts, feedback contributions, and email dispatch counts into a single printable and exportable summary per admin. Reports are exportable as CSV files for use in payroll systems, performance review documentation, or external analytics tools. The module complements the live Performance Scores leaderboard by providing a historical, archivable snapshot rather than a real-time view.

Module 26: Birthday Reminders (All Users – HR & Operations)

The Birthday Reminders module surfaces upcoming and today's birthdays for all team members including admins and employees, sourcing birth dates from Employee Management and admin profile records. Today's birthdays are highlighted with a special celebration banner at the top of the module. Upcoming birthdays within the next 7 days are listed with name, designation, and countdown in days, ensuring no team occasion goes unnoticed. The

module also aggregates birthday data into the HR Dashboard for HR personnel who coordinate birthday recognition activities.

Module 27: Email Management (Super Admin — Communication)

The Email Management module enables authenticated Super Admins to compose and dispatch branded professional emails directly from the portal using the company's custom sender domain @admin.thrylos.in, powered by the Brevo transactional email API with verified DKIM and SPF DNS records for maximum inbox deliverability. The module enforces configurable daily send quotas per admin to prevent misuse, stores a complete sent-email log with recipient, subject, sending admin, timestamp, and delivery status, and feeds summary statistics (emails sent today, this month) into Admin Reports. This module is suitable for client outreach, formal communications, and team announcements that require the company's official domain.

Module 28: Quick Notes (All Users — Productivity)

Quick Notes provides each admin with a private, personal sticky-note workspace for capturing ideas, tasks, reminders, and reference information that should not be shared with the team. Notes can be created, color-coded with a palette of six colors, pinned to the top for priority visibility, edited inline, and deleted. Notes are strictly personal — Supabase RLS policies ensure that no admin, including Super Admins, can read another admin's notes. A keyword search filter lets admins quickly locate specific notes within large note collections.

Module 29: Company Directory (All Users — HR & Operations)

The Company Directory provides a searchable, filterable internal people directory covering all admins and employees. Each profile card displays the person's name, designation, department, email address, phone number (if shared), and profile picture. The directory supports search by name, filter by department, and filter by role level. It serves as the primary onboarding resource for new team members orienting themselves within the organization and as the reference for cross-department communication. Contact details are sourced directly from Employee Management and admin profile records, ensuring the directory is always current.

Module 30: Meeting Notes (All Users — Project Management)

Meeting Notes is a collaborative documentation module for recording, storing, and sharing structured meeting minutes. Each entry captures the meeting date, attendees (tagged from the Company Directory), agenda items, key discussion points, decisions made, and action items with assigned owners and due dates. Notes are archived chronologically and searchable by date, meeting type, or keyword. The action item tracker within each note entry integrates with the Task Board — action items can be converted into Task Board cards with a single click, ensuring no follow-up is lost after a meeting.

Module 31: Expense Claims (All Users — Finance)

The Expense Claims module provides a streamlined reimbursement management workflow. Employees and admins submit claims specifying the expense category (travel, equipment, food, accommodation, or miscellaneous), amount, date, description, and a supporting receipt uploaded through Supabase Storage. Claims are routed through a configurable approval workflow where the final approver is typically a Super Admin or Finance-designated admin. The module tracks total claims submitted, pending approval count, approved total

amount, and complete approval history per user, providing the financial accountability required for internal reimbursement processes.

Module 32: Company Links (All Users – Productivity)

Company Links is a curated, categorized bookmarks repository providing the entire team with one-click access to frequently used company tools, internal documents, management portals, analytics dashboards, and external resources. Links are organized into categories such as Design Tools, Development Resources, HR Portals, Analytics Dashboards, Social Media Management, and Client Portals, flagged as important where critical, and searchable by name or category. This module eliminates the common productivity friction of team members needing to ask colleagues for specific tool URLs or rummaging through email threads for shared resource links.

Module 33: Esports Players (Super Admin – Esports Operations)

The Esports Players module is Thrylos India's dedicated player registry and tournament operations tool, directly supporting the company's core esports business vertical. It allows registering tournament participants with gaming handles, game titles, ranks, performance statistics, and payment records. The module tracks total registered users, cumulative tournament registration revenue, and individual player activity history. Bulk CSV import support allows tournament organizers to register hundreds of players from registration forms in a single operation. Player data integrates with the Analytics Dashboard for revenue reporting and with the Payment Verification module for transaction reconciliation.

Module 34: Social Media Analytics (Super Admin – Dashboard & Analytics)

The Social Media Analytics module provides cross-platform content and community performance tracking for Thrylos India's social media presence. It tracks total posts published per platform, followers gained over configurable time periods, engagement metrics (likes, comments, shares, saves), collaboration requests received from content creators or brand partners, and audience response data. Recharts visualizations provide period-over-period growth comparisons, platform-wise follower distribution, and engagement rate trend lines. This module supports the content team's performance reviews and the company's social media strategy decisions with quantitative data rather than platform-native analytics.

Module 35: Tech Work Dashboard (Super Admin – Project Management)

The Tech Work Dashboard is a developer productivity and work-logging module enabling the technical team to submit structured daily work logs. Each log entry records pages or features created, bugs identified, bugs fixed, new features added, hours worked, and any blockers. Aggregated metrics display total work pages logged across the team, total hours, individual contribution rates, and bug resolution velocities. The module integrates with Performance Scores, where tech work log data contributes to the weighted performance calculation for technical team members.

Module 36: Content Work Dashboard (Super Admin – Dashboard & Analytics)

The Content Work Dashboard tracks content production output for the content and media team. Each entry records content pieces created by type (social media posts, reels, YouTube videos, blog articles, graphic banners, event posters), the platforms published on, associated metrics such as reach and engagement for tracked content, and the responsible content creator. The module tracks total content items created and active platforms per period. It feeds directly into both the Performance Scores engine (as the content creation count metric for non-technical team members) and the Social Media Analytics module for cross-referencing published content with engagement outcomes.

Chapter 7

Implementation and Experimental Setup

7.1 Real-Time Team Chat System

The Team Chat module was the most technically demanding component of the entire project, representing a significant departure from standard CRUD operations into real-time distributed systems territory. The implementation is built entirely on Supabase Realtime's WebSocket pub-sub infrastructure, with no dependency on third-party chat SDKs. This decision kept all message data within the company's own PostgreSQL database, subject to the same RLS policies and audit logging as all other data in the system.

The WebSocket subscription architecture was carefully designed to avoid the message duplication bug encountered during integration (detailed in Chapter 10). The final architecture uses a single shared subscription context provider mounted at the application root, with an internal event emitter routing incoming Supabase Realtime events to the relevant component subscribers. This prevents the scenario where multiple components independently subscribing to the same Supabase channel receive and render duplicate messages.

File attachments in chat are handled by uploading the file to a dedicated Supabase Storage bucket and inserting the storage path into the message record rather than storing binary data in the database. Signed URLs with configurable expiry times are generated on demand when a recipient renders the message, ensuring attachments are accessible only to authenticated users with appropriate role access.

7.2 Performance Scoring Engine

The Performance Scores engine represents a non-trivial algorithmic implementation: a multi-source, multi-weighted scoring model that aggregates data from six independent modules and recalculates in real time. The scoring formula is: $\text{TotalScore} = (\text{AttendancePct} \times W1) + (\text{WorkLogHours} \times W2) + (\text{BugsFixed} \times W3) + (\text{FeaturesDelivered} \times W4) + (\text{ContentCreated} \times W5) + (\text{TaskCompletionRate} \times W6) + (\text{StandupConsistency} \times W7)$, where each weight W_i is configurable by Super Admins through the portal interface.

Recalculation is triggered by Supabase Realtime events from each of the six source modules rather than by a scheduled batch job. This means that as soon as a team member marks an attendance record, submits a work log, closes a task, or posts a standup, the entire leaderboard recalculates within milliseconds. The computational cost is acceptable because the calculation involves at most a few hundred records for a typical team size.

7.3 Audit Log Infrastructure

The Audit Logs module functions as a forensic record of all significant system activity. Implementing its insert-only requirement required a specific Supabase RLS policy pattern: the table has an INSERT policy allowing authenticated admins to create new records, but DELETE and UPDATE policies are completely absent (not set to FALSE, but truly absent), which in Supabase's permissive RLS model defaults to denying those operations for all users. This means that even a user with the Supabase service role key, if exposed to a Super Admin through a mistaken configuration change, cannot delete audit records through the REST API.

The audit log schema captures eleven fields per record: `admin_id` (UUID, foreign key to admin profiles), `action_type` (enum: login, logout, create, update, delete, export, view_sensitive, config_change, auth_failure), `module` (string), `record_id` (nullable UUID of the affected record), `timestamp` (timestampz with millisecond precision), `ip_address`, `city`, `region`, `country`, `isp`, and `gps_coordinates` (PostGIS point type). The breadth of this schema ensures that any future security investigation can reconstruct the exact sequence of events for any time period.

7.4 Supabase Edge Functions Architecture

Three critical backend operations run as Supabase Edge Functions on the Deno TypeScript runtime: OTP generation and storage, Brevo email dispatch, and quota enforcement. Edge Functions were chosen over a Node.js Express server for several reasons: they execute within Supabase's globally distributed edge network (low latency regardless of admin location), they share the same authenticated Supabase context as the database (no separate service-to-service authentication), and Supabase Secrets management keeps sensitive credentials like the Brevo API key completely out of the frontend bundle and the GitHub repository.

The OTP generation function uses Node.js's `crypto.randomInt()` equivalent in Deno to generate a cryptographically random 6-digit integer, salts it with a per-admin secret stored in Supabase Secrets, and stores the HMAC hash (not the raw OTP) in a Supabase table alongside the expiry timestamp and consumed flag. The raw OTP is only included in the Brevo email dispatch call and never stored in plaintext anywhere in the system.

7.5 Bulk Import and Data Validation

The Bulk Upload and Import module solves a specific operational problem: tournament organizers regularly need to import 200–1000 player registration records that have been collected through external forms, without having direct database access. The module's client-side validation using Papa Parse catches data quality issues before any Supabase API call is made. The validation schema for each import type (player registrations, employee batches, inventory) is stored as a JSON schema definition within the codebase and can be extended for new import types without modifying the core module.

The performance optimization from individual record insertion (which timed out for 500+ row files on Supabase's free tier) to bulk upsert using Supabase's `supabase.from().upsert()` API with arrays of records reduced import time for 1,000-row files from 45 seconds to under 2 seconds, making the feature actually usable for production tournament operations.

7.6 Dashboard Interface and UX Design

The portal's user interface is built on an information density without cognitive overload design philosophy. Every screen surfaces the most critical information at a glance while providing depth on demand through progressive disclosure: expandable sections, collapsible panels, and drill-down navigation. The main dashboard's five-zone layout (welcome banner, tasks widget, Super Admin accordion, stats row, module grid) was iterated through three design versions based on feedback from Thrylos India's management team before settling on the final structure.

Each of the 36 module cards on the dashboard follows a consistent information architecture: a domain emoji or icon, an access level badge, a live status badge where real-time data is active, the module name in bold, a one-sentence functional description, two to three live-fetched metric fields from Supabase, and a primary Open action button. The card grid transitions from 4 columns on wide desktop screens to 2 columns on tablets and single column on mobile through Tailwind CSS responsive breakpoints. Framer Motion staggered entrance animations cascade the 36 cards in sequence on dashboard load, creating the perception of fast rendering even when Supabase data fetches are still resolving.

Chapter 8

Results and Discussion

8.1 Delivery Outcomes

The project delivered all eight originally defined objectives, plus six additional modules beyond the minimum 30 specified. The final deployed system at admin.thrylos.in contains 36 fully operational production modules, four role-based access levels, an enterprise-grade two-factor authentication system with geolocation logging, a real-time team chat platform, a configurable performance scoring engine, and comprehensive audit logging infrastructure. The system is live, serving real users, and processing real operational data at Thrylos India.

From a code quality perspective, the project maintained TypeScript strict mode compliance throughout the entire 50,000+ line codebase with zero type errors at the time of production deployment. All database operations are protected by RLS policies. The Vercel CI/CD pipeline has successfully executed over 40 automatic production deployments from main branch pushes without manual intervention or deployment failures. Supabase Realtime subscriptions across the five real-time modules have operated without reconnection issues or message duplication (following the architecture refactor documented in Chapter 10).

8.2 Student Team and Individual Contributions

Ishan Jha (2401010022) — Project Lead & Principal Architect

Ishan Jha served as Project Lead and Principal Architect. He designed the overall system architecture, PostgreSQL database schema, and module dependency map. He built the core authentication flow including OTP generation logic, Brevo API integration in Edge Functions, geolocation session logging, and JWT session management. He developed the main Dashboard, Overview Stats, Audit Logs, Admin Management module, and the global navigation and React Router routing infrastructure.

Utsav Ratan (2401010046) — Frontend Developer

Utsav Ratan served as a core Frontend Developer responsible for building and integrating client-side module interfaces across the portal. He implemented React 18 component architectures for multiple modules, handled routing integrations, and worked on state management patterns ensuring smooth cross-module data flows. He collaborated on the responsive layout system and contributed to the consistent implementation of UI components using Tailwind CSS and shaden/ui across the application.

Rashmi Jha (2401010166) — UI/UX Designer

Rashmi Jha served as the UI/UX Designer responsible for the complete visual design language and user experience architecture of the portal. She created wireframes, defined the component library setup, and established design token definitions and animation specifications. She designed the information hierarchy for all 36 module cards, the dark header and branded colour scheme, and the overall visual identity of the interface. She drove mobile responsiveness validation across all device form factors and ensured the portal met baseline accessibility standards throughout all modules.

Prem Kumar (2401010243) — Frontend Developer

Prem Kumar served as a Frontend Developer responsible for the complete suite of HR and Operations module interfaces: Employee Management, Leave Management, Internship Tracker, HR Dashboard, Holiday Calendar, Birthday Reminders, Expense Claims, and Career Applications. He designed and implemented the leave approval workflow UI and integrated all HR data pipelines with the Performance Scores engine on the frontend, ensuring that attendance and leave data correctly feeds into performance calculations.

Vicky Jha (2401010248) — Backend Developer

Vicky Jha served as the primary Backend Developer responsible for server-side logic, API integrations, and data pipeline architecture across the portal. He built Supabase Edge Functions powering esports operations, developed the backend logic for the Attendance Tracker, Daily Standups, Content Work Dashboard, and Tech Work Dashboard modules, and implemented the Bulk Upload CSV processing pipeline. He integrated esports data flows with the Analytics Dashboard revenue reporting and Performance Scores modules, and managed backend query optimization across high-volume data operations.

Vivek Rai (2401010303) — Database Engineer

Vivek Rai served as the Database Engineer responsible for designing all Supabase PostgreSQL table schemas, relationships, and Row-Level Security policies. He authored comprehensive RLS policies for all 36 modules ensuring defence-in-depth access control at the database engine level independent of application logic. He designed the normalized schema covering HR, finance, communication, and esports data domains, managed all Supabase Storage bucket configurations and access policies, and oversaw the audit log storage pipeline infrastructure.

Chapter 9

Conclusion and Future Work

9.1 Conclusion

The Thrylos India Admin Management Portal represents a genuinely exceptional outcome for an industry project undertaken by first-year undergraduate B.Tech students. Over five months, the six-member team designed, built, tested, and deployed a 36-module enterprise-grade web application that is currently live at admin.thrylos.in, serving real users, and underpinning the day-to-day operations of Thrylos India as its authoritative internal management platform.

The project demonstrates that with deliberate technology choices, structured teamwork, and a commitment to production-quality standards over academic shortcuts, student developers can deliver software systems that are indistinguishable in quality, functionality, and reliability from those produced by professional engineering teams. The portal's authentication architecture, real-time WebSocket infrastructure, role-based access control enforced at both the application and database layers, and comprehensive 36-module functional coverage all reflect genuine engineering maturity developed through the pressure of building for real users.

Beyond the technical deliverable, this project provided the student team with professional development experiences that no classroom exercise could replicate: navigating real client requirements with evolving scope, debugging production issues affecting real users, managing a multi-person codebase over an extended timeline with structured Git workflows, and ultimately taking end-to-end ownership of a live system. These experiences constitute permanent career differentiators for each team member.

9.2 Future Scope and Planned Enhancements

Several enhancements have been identified for future development phases. A native mobile application using React Native would enable on-the-go attendance marking, leave approvals, and real-time chat without requiring a browser session, significantly improving usability for field-based team members. An AI-powered admin assistant integrated via the Claude or GPT API would allow natural language queries within the portal: admins could ask questions like 'Show me attendance for last week' or 'Who has pending expense claims over five thousand rupees' and receive instant data-driven answers without navigating to specific modules.

Advanced predictive analytics would extend the Analytics Dashboard with time-series models forecasting attendance trends, revenue trajectories, and employee performance projections based on historical data already stored in Supabase. A multi-company architecture refactoring would transform the platform into a multi-tenant SaaS product, allowing Thrylos India to offer the portal as a white-label service to other esports organizations and digital companies, creating a revenue stream from the engineering investment made during this project.

A full payroll processing module integrating with the Leave Management, Attendance Tracker, and Employee Management modules would automate salary computation, deduction handling, and payslip generation, eliminating the remaining manual HR process. An API rate limiting dashboard would give Super Admins real-time visibility into Brevo API consumption, Supabase Edge Function invocation counts, and storage bandwidth, with configurable alerting thresholds. Automated report scheduling would allow Super Admins to configure weekly or monthly Admin Reports to be generated and emailed automatically to designated recipients.

Learning Outcomes and Skill Acquisition

This industry project provided each team member with technical and professional learning that is qualitatively different from what standard academic coursework or laboratory exercises can deliver. The requirement to build a system that would be used by real people, with real consequences for failures and real gratitude for well-designed features, created a quality standard and personal accountability that closely mirrors a professional software engineering environment.

Technical Skills Acquired

In React 18 and TypeScript, the team developed custom hooks for data fetching and state abstraction, Context API patterns for global authentication and role state management, compound component patterns for complex UI elements, and TypeScript generics, discriminated unions, and strict null checks across a 50,000+ line codebase. In Supabase and PostgreSQL, the team learned normalized schema design with foreign key relationships, Row-Level Security policy authoring for complex access control scenarios, Edge Function development in Deno TypeScript, Realtime subscription patterns for multi-client state synchronization, and storage bucket access policy management.

In API integration, the team implemented the Brevo transactional email API with DKIM/SPF domain verification, IP geolocation microservice integration, Browser Geolocation API with graceful fallback patterns, and file upload APIs with progress tracking. In frontend architecture, the team built feature-based folder structures for large React applications, React Router v6 with nested routes and protected route wrappers, Framer Motion production animation orchestration, and Tailwind CSS design token systems. In data visualization with Recharts, the team built composable charts with responsive containers, accessible ARIA labels, hover tooltips, and composite multi-axis configurations. In deployment and CI/CD, the team configured Vercel production pipelines, custom domain DNS with A records and CNAME, production versus preview environment branching, and build optimization with bundle analysis.

Professional Skills Developed

The team developed agile development workflow skills through five months of sprint planning, daily standups, iterative feature delivery, and retrospective reviews. Requirements engineering skills were built by translating Thylos India management's operational needs and evolving feedback into precise technical specifications. Cross-functional collaboration was practiced by coordinating six developers with distinct roles while maintaining a consistent codebase and design language. Technical communication skills were developed through writing GitHub documentation, API documentation, and this formal academic report. Client relationship management was practiced through regular progress presentations, feature demonstrations, and professional handling of scope changes. Code review practices were built through structured pull request workflows with constructive peer feedback.

Chapter 10

Challenges Faced and Solutions Implemented

Building a production-grade system serving real users across 36 modules inevitably surfaces engineering challenges that do not appear in controlled academic exercises. This chapter documents the eight most significant challenges encountered during the project and the technical solutions implemented.

Challenge 1: OTP Emails Classified as Spam

The initial `@admin.thrylos.in` sender domain lacked DKIM public key and SPF TXT records in the domain's DNS configuration, causing major email providers including Gmail, Outlook, and Yahoo to classify outgoing OTP emails as potential phishing attempts. This made the authentication system functionally unusable for approximately 40% of test admins during the first integration test. The solution involved configuring the proper DKIM public key record and SPF TXT record in the domain registrar's DNS settings and verifying the sending domain in Brevo's dashboard. After DNS propagation, OTP email deliverability reached 100% across all tested providers.

Challenge 2: Supabase Realtime Message Duplication in Team Chat

Messages were appearing twice in the Team Chat UI due to a dual-subscription conflict: one Supabase Realtime subscription from the chat component and a second subscription from the global notification listener were both processing incoming message events independently and passing them to React state, triggering two separate renders each adding the message to the displayed list. The solution was to refactor the subscription architecture to use a single shared subscription context provider mounted at the app root, with an internal pub-sub event emitter routing incoming events to relevant component subscribers. This eliminated the duplicate subscription and the resulting duplication.

Challenge 3: RLS Policies Blocking Legitimate Super Admin Access

Several early Row-Level Security policies were written with an `auth.uid() = creator_id` condition without a Super Admin bypass clause. This prevented Super Admins from reading records created by other admins — for example, a Super Admin could not see another admin's attendance records because the policy only allowed the record creator to read their own data. The fix revised all affected policies to include an OR clause: `(auth.uid() = creator_id) OR (get_admin_role(auth.uid()) = 'super_admin')`, using a Supabase database function to retrieve the requesting admin's role. All policies were regression-tested after modification to verify both the restriction and the bypass behavior.

Challenge 4: Framer Motion Layout Animations Causing Scroll Jumps

`AnimatePresence` layout animations on the 36-module card grid were causing the page scroll position to jump unexpectedly when cards entered or exited the DOM, because layout animations trigger a full layout recalculation

that repositions all sibling elements simultaneously. The solution replaced `layout={true}` animations (which trigger full layout recalculation) with `translateY` and `opacity` entrance animations using Framer Motion's variants API. This preserved the visual cascade entrance effect while eliminating the scroll interference entirely.

Challenge 5: Browser Geolocation API Blocked on Some Devices

Several admins had geolocation permissions blocked at the browser or operating system level, causing login to fail entirely because the original authentication flow required successful GPS coordinate capture before creating the Supabase session. This was a usability regression: admins using privacy-focused browser configurations or corporate devices with location services disabled were locked out. The solution restructured the auth flow to create the Supabase session first, then attempt geolocation as a secondary step with a 5-second timeout. If geolocation fails or times out, the session is still established and the login is logged with IP-only location data as a graceful fallback.

Challenge 6: Large CSV Imports Timing Out

Bulk imports of 500+ row CSV files were timing out due to Supabase's connection pool limits combined with the original approach of inserting records individually in a JavaScript for-loop, which generated one database round-trip per record. For 500 records this meant 500 sequential Supabase API calls, exceeding the connection pool timeout. The solution refactored the import pipeline to use Supabase's `supabase.from().upsert()` bulk insert API, which batches all validated records into a single PostgreSQL upsert statement. Import time for 1,000-row files dropped from 45 seconds to under 2 seconds.

Challenge 7: TypeScript Type Errors Across Module Boundaries

As the module count grew past 15 and more team members were building modules simultaneously, shared data types such as `User`, `Admin`, `Task`, `Leave`, and `AttendanceRecord` were being defined inconsistently in multiple files. TypeScript's strict mode raised hundreds of type mismatch errors during integration when one module expected a field that another module's definition did not include. The solution created a centralized `/types` directory with a single source-of-truth file for all shared interfaces and type definitions. All 36 modules were updated to import from this central file. TypeScript strict mode was enforced at the `tsconfig` level to prevent future drift.

Challenge 8: Mobile Layout Breaking on Small Screen Widths

The original grid and table layouts were designed desktop-first and broke significantly on screens narrower than 375px, with tables overflowing their containers and card grids collapsing poorly. The solution adopted a mobile-first approach for all new layouts going forward and retroactively fixed existing layouts: tables were wrapped in `overflow-x-auto` containers to enable horizontal scrolling on small screens, the module card grid was converted to CSS Grid with auto-fill columns using Tailwind's responsive breakpoint classes (`grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4`), and all fixed pixel spacing was converted to responsive Tailwind utilities.

Annexure I: Plagiarism Declaration Certificate

Title of Work: Thrylos India Admin Management Portal — Industry Project Report

Submitted By: Utsav Ratan (2401010046), Ishan Jha (2401010022), Rashmi Jha (2401010166), Prem Kumar (2401010243), Vicky Jha (2401010248), Vivek Rai (2401010303)

Institution: K. R. Mangalam University, Gurugram

Department: Computer Science and Engineering

Date of Submission: April 2026

We hereby declare that the work entitled “Thrylos India Admin Management Portal” submitted for academic evaluation and research purposes is our original work. We confirm that:

1. We have acknowledged and properly cited all sources, references, libraries, frameworks, and third-party services included in this work.
2. This work does not contain any material previously published, written, or prepared by another person, except where due acknowledgment has been made.
3. We understand that plagiarism is an academic offense and a violation of research ethics. Any breach of this declaration may lead to disciplinary action as per university policies.
4. We have maintained academic integrity throughout this work and in the code, architecture, and documentation of the delivered system.
5. We affirm that the submitted work has a plagiarism similarity score of less than 10% and is free from AI-generated content substituting original research or analysis.

Signature of Students: Utsav

Prem
Rashmi
Vicky
Vivek
Ishan

Date: 12/04/2026

References

The following technologies, libraries, frameworks, and services were used in the development of the Thrylos India Admin Management Portal and are acknowledged here:

1. React 18 Documentation and Migration Guide — react.dev. Meta Platforms, Inc., 2024.
 2. TypeScript Handbook — typescriptlang.org. Microsoft Corporation, 2024.
 3. Supabase Documentation: PostgreSQL, Row-Level Security, Edge Functions, and Realtime — supabase.com/docs. Supabase, Inc., 2024.
 4. Tailwind CSS Documentation — tailwindcss.com. Tailwind Labs, Inc., 2024.
 5. shadcn/ui Component Library — ui.shadcn.com. shadcn, 2024.
 6. Framer Motion Documentation — framer.com/motion. Framer B.V., 2024.
 7. Recharts Documentation — recharts.org. Recharts Group, 2024.
 8. Brevo (Sendinblue) Transactional Email API Documentation — developers.brevo.com. Brevo SAS, 2024.
 9. Vercel Deployment Documentation — vercel.com/docs. Vercel, Inc., 2024.
 10. React Router v6 Documentation — reactrouter.com. Remix Software, Inc., 2024.
 11. Papa Parse CSV Parsing Library Documentation — papaparse.com. Matthew Holt, 2024.
 12. NIST Special Publication 800-63B: Digital Identity Guidelines, Authentication and Lifecycle Management. National Institute of Standards and Technology, 2017 (updated 2022).
 13. Deno Runtime Documentation — deno.land/manual. Deno Land Inc., 2024.
 14. PostgreSQL 15 Documentation: Row Security Policies — postgresql.org. The PostgreSQL Global Development Group, 2024.
-